

FocusAid — Technical Brief

Built in 5 hours by four developers working in parallel on one repo.

Stack: Next.js 16 (App Router, Turbopack) · React 19 · TypeScript · Tailwind v4 · MediaPipe Tasks-Vision · transformers.js · Anthropic SDK

ONE CLOCK

`src/app/page.tsx` owns a single `currentTime`; transcript, captions, attention timeline and catch-up all derive from it. That, a frozen shared type contract, and ownership split by directory is what let four people build independently and still agree on what “2:14” means.

ATTENTION — IN-BROWSER, NOTHING UPLOADED

MediaPipe Face Landmarker: 478 landmarks, 52 blendshapes, facial transformation matrices. Gaze-away from iris position in the eye socket; confusion from brow-lowerer blendshapes; head alignment from matrix element `d[10]` — index (2,2), the same cell under both row- and column-major layouts, so the read is transpose-invariant — divided by axis length to cancel scale. Events carry a 4-second floor so a glance never alerts, and **an unobserved second ends a run rather than extending it**: the system never claims the student was away during time it did not watch.

TRANSCRIPTION

Browser `SpeechRecognition`, continuous and self-restarting after silence, with phrases stamped from when speech *started* rather than when the engine committed them, so lines track the clock instead of lagging it. Captions are *derived* from the active transcript and cannot drift from it.

AI — THREE TIERS, ALWAYS DEGRADING

1	Hosted model · <code>/api/summary</code>	Structured JSON against a fixed schema; 6s timeout, no retries. Falls to tier 3 on any error, refusal or malformed reply.
2	On-device · SmolLM2-360M q4f16, WebGPU	<code>transformers.js</code> in a Web Worker , so generation never stutters caption rendering or the tracker. Opt-in, cached, offline after first load.
3	Deterministic · from the transcript	Synchronous, renders the instant the student taps. Tiers 1–2 are upgrades layered on top; nothing stands between the request and an answer.

The on-device model selects; it never writes.

Asked to rewrite passages, SmolLM2-360M was faithful on **0 of 6** test windows — it inverted CO₂ for oxygen, carbon for sugar, light-dependent for light-independent. Qwen2.5-0.5B at 460 MB was no better. Because our user cannot check generated text against audio, the model is now constrained to picking one of the teacher's own sentences, and every reply is matched against the source before display; non-verbatim output is discarded. Measured: **5/6** windows yield a verified key point, **5/5** verifier unit checks including a planted hallucination. Diagrams stay derived from the transcript, so no model can invent a relationship between concepts.

ACCESSIBILITY

ARIA tabs with roving tabindex and arrow/Home/End navigation; focus rings on every control; `prefers-reduced-motion` honoured globally; high-contrast captions; fuzzy transcript search; notes persisted via `useSyncExternalStore` for an SSR-safe read with no hydration flash.